FARNAM MANSOURI

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RESEARCH INTERESTS

Statistical Learning Theory	Language Learning Theory	Optimization
Reinforcement Learning Theory	Machine Teaching	Deep Learning

EDUCATION

University of Waterloo *Jan 2023 – Dec 2027 (Expected)*
Ph.D. in Computer Science
Advised by Shai Ben-David and Sandra Zilles

University of Toronto *Sep 2020 – Aug 2022*
M.Sc. in Computer Science
Advised by Murat Erdogdu and Amir-massoud Farahmand

Sharif University of Technology *Sep 2015 – Aug 2020*
B.Sc. in Software Engineering with a minor in Mathematics

PUBLICATIONS

Refereed

[P1] Surprises in Proper Positive-Only Learning
S. Ben-David, **F. Mansouri**, A. Mehrotra, M. Zampetakis (alphabetical order)
The 40th Annual Conference on Neural Information Processing Systems (NeurIPS), 2026;
Oral Presentation, Top 0.4% of Submissions

[P2] Near-Optimal Sample Complexity of Deterministic Agnostic PAC Learning
S. Ben-David, S. Hanneke, **F. Mansouri**, A. Shaeiri (alphabetical order)
The 40th Annual Conference on Neural Information Processing Systems (NeurIPS), 2026

[P3] Learning with Multiple Correct Answers – Regret Bounds under Different Feedback Models
A. F. Pour*, **F. Mansouri***, S. Ben-David
The 40th Annual Conference on Neural Information Processing Systems (NeurIPS), 2026

[P4] Active learning from positive and unlabeled examples
F. Mansouri, S. Zilles, S. Ben-David
The 40th Annual Conference on Neural Information Processing Systems (NeurIPS), 2026

[P5] The Computational Complexity of Almost Stable Clustering with Penalties
F. Mansouri, S. Zilles, K. Khodamoradi
The 35th International Joint Conference on Artificial Intelligence (IJCAI), 2026

[P6] Learning from positive and unlabeled examples—Finite size sample bounds
F. Mansouri, S. Ben-David
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

[P7] Formal Models of Active Learning from Contrastive Examples
F. Mansouri, H. U. Simon, Y. Chen, A. Singla, S. Zilles
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

[P8] On Batch Teaching with Sample Complexity Bounded by VCD
F. Mansouri, H. U. Simon, A. Singla, S. Zilles
The 36th Conference on Neural Information Processing Systems (NeurIPS), 2022; **Spotlight Paper**

[P9] Deep Reinforcement Learning for Online Control of Stochastic Partial Differential Equations
E. Pirmorad*, F. Khoshbakhtian*, **F. Mansouri***, A. M. Farahmand
NeurIPS Workshop on the Symbiosis of Deep Learning and Differential Equations (DLDE), 2021; **Spotlight paper**

[P10] ChOracle: A Unified Statistical Framework for Churn Prediction
A. Khodadadi, A. Hosseini, E. Pajouheshgar, **F. Mansouri**, H. R. Rabiee
IEEE Transactions on Knowledge and Data Engineering (IEEE TKDE), 2020

[P11] Understanding the Power and Limitations of Teaching with Imperfect Knowledge
R. Devidze*, **F. Mansouri***, L. Haug, Y. Chen, A. Singla
The 29th International Joint Conference on Artificial Intelligence (IJCAI), 2020

[P12] Preference-Based Batch and Sequential Teaching: Towards a Unified View of Models
F. Mansouri, Y. Chen, A. Vartanian, X. Zhu, A. Singla
The 33rd Conference on Neural Information Processing Systems (NeurIPS), 2019

Preprints and Manuscripts

[M1] PAC Learning with Multiple Correct Answers
S. Hanneke, **F. Mansouri**, A. F. Pour, A. Shaeiri, K. Wang (alphabetical order)
Submitted to the 59th Annual ACM Symposium on Theory of Computing (STOC), 2027

[M2] Deterministic Proper Positive-Only Learning is Independent of ZFC
S. Ben-David, **F. Mansouri**, A. Mehrotra (alphabetical order)
Submitted to the 38th International Conference on Algorithmic Learning Theory (ALT), 2027

[M3] Query Feedback and Teaching for Stable and Certifiable Language Generation
F. Mansouri, A. F. Pour, G. Velegkas, S. Zilles (alphabetical order)
Submitted to the 38th International Conference on Algorithmic Learning Theory (ALT), 2027

[M4] Sample Complexity of Learning from Hidden Thinkers
S. Ben-David, N. Joshi, **F. Mansouri**, A. F. Pour (alphabetical order)
In Preparation for the 40th Annual Conference on Learning Theory (COLT), 2027

[M5] On the Convergence and Approximation of Policy Gradient Methods for Spectral Measures
F. Mansouri
M.Sc. Thesis, 2022

[M6] Learning Half-Spaces from Perturbed Contrastive Examples
A. A. R. Ravari*, **F. Mansouri***, Y. Chen, V. Ivarson, A. Singla, S. Zilles
arXiv:2602.02080, 2026

[M7] Preference-Based Batch and Sequential Teaching
F. Mansouri, Y. Chen, A. Vartanian, X. Zhu, A. Singla
arXiv:2010.10012, 2020

*equal contributions.

PROFESSIONAL EXPERIENCE

RBC Borealis *Oct 2026 – Feb 2027 (Expected)*
Machine Learning Research Internship *Toronto, Canada*
Research on statistical complexity of modern LLMs.

Max Planck Institute for Software Systems *July 2018 – Sep 2019*
Research Internship *Saarbruecken, Germany*
Under the supervision of Adish Singla, Machine Teaching Group.

Yektanet *June 2017 – Sep 2017*
AI Engineer *Tehran, Iran*
Developed a framework to classify Persian web-page content into 12 categories. Yektanet is the largest recommendation system platform in Iran.

PROGRAMMING SKILLS

Python R Java C/C++ PyTorch TensorFlow scikit-learn PostgreSQL

AWARDS AND HONORS

David R. Cheriton Graduate Scholarship *2026–2028*
20,000 CAD

NSERC Canada Graduate Scholarship-Doctoral (CGS-D) *2024–2027*
120,000 CAD

David R. Cheriton Graduate Scholarship *2023–2025*
20,000 CAD

Ontario Graduate Scholarship *2020–2021*
15,000 CAD

Vector Institute Scholarship in Artificial Intelligence *2020–2021*
17,500 CAD

Iran National Mathematics Olympiad *2014*
Silver medal; among the top 40 students in Iran, out of more than 30,000 initial participants.

TALKS

Surprises in Proper Positive-Only Learning *Dec 2026*
Conference on Neural Information Processing Systems (NeurIPS) *Sydney, Australia*

The Computational Complexity of Almost Stable Clustering with Penalties *Aug 2026*
International Joint Conference on Artificial Intelligence (IJCAI) *Bremen, Germany*

On Batch Teaching with Sample Complexity Bounded by VCD *Dec 2023*
Winter meeting of the Canadian Mathematical Society *Montreal, Canada*
Special session: "Current trends in matrices, graphs and computing".

PROFESSIONAL SERVICES

Conference Program Committee Member/Reviewer for **NeurIPS** (2021, 2022, 2023, 2025, 2026), **ICML** (2023, 2024, 2025, 2026), **AAAI** (2023), **ICLR** (2024)

TEACHING

Teaching Assistant Experiences

Vector Institute *Toronto, Canada*
- Causal Inference workshop *Spring 2023*

University of Waterloo *Waterloo, Canada*
- Data Structures and Data Management *Winter 2025, Spring 2025*
- Algorithmic Problem Solving *Spring 2024*
- Data Types and Structures *Fall 2024*
- Logic and Computation *Spring 2023, Fall 2024 (IA), Winter 2024, Fall 2025, Fall 2026*
- Elementary Algorithm Design and Data Abstraction *Winter 2023, Winter 2026*

University of Toronto *Toronto, Canada*
- Machine Learning for Healthcare* *Fall 2021*
- Minimizing Expectations* *Winter 2021*
- Introduction to Machine Learning *Fall 2020*

Sharif University of Technology *Tehran, Iran*
- Deep Learning* *Spring 2019*
- Machine Learning* *Fall 2018*
- Linear Algebra *Fall 2018*
- Design of Algorithms *Spring 2017*
- Engineering Probability and Statistics *Fall 2016, Spring 2016*

* Graduate Courses

Tutoring *2015–2016*

Teaching Combinatorics for high school students, preparing for Iran National Mathematics competition at Helli 1 High School. *Tehran, Iran.*